

8086 Programs

9. Write an assembly language program for adding two 16-bit data using 8086 processor.

The screenshot shows the emu8086 emulator interface. The 'original source' window contains the following assembly code:

```

101: NUM1 DW 1234H
102: NUM2 DW 1111H
103: RES DW ?
104: DATA ENDS
105:
106: CODE SEGMENT
107: ASSUME CS:CODE, DS:DATA
108:
109: START:
110: MOV AX, DATA
111: MOV DS, AX
112:
113: MOV AX, NUM1
114: ADD AX, NUM2
115: MOV RES, AX
116:
117: MOV AX, 4CH
118: INT 21H
119:
120: CODE ENDS

```

The 'registers' window shows the state of the 8086 registers. The AX register contains 0000, and the DS register contains 0000. The 'DATA' window shows the memory contents, with NUM1 at 1234H and NUM2 at 1111H.

10. Write an assembly language program for subtracting two 16-bit data using 8086 processor.

The screenshot shows the emu8086 emulator interface. The 'original source' window contains the following assembly code:

```

101: DATA SEGMENT
102: NUM1 DW 1234H
103: NUM2 DW 1111H
104: RES DW ?
105: DATA ENDS
106:
107: CODE SEGMENT
108: ASSUME CS:CODE, DS:DATA
109:
110: START:
111: MOV AX, DS
112: MOV DS, NUM1
113: MOV RES, AX
114: SUB AX, NUM2
115: MOV RES, AX
116:
117: MOV AX, 4CH
118: INT 21H
119:
120: CODE ENDS

```

The 'registers' window shows the state of the 8086 registers. The AX register contains 0000, and the DS register contains 0000. The 'DATA' window shows the memory contents, with NUM1 at 1234H and NUM2 at 1111H.

11. Write an assembly language program for multiplying two 16-bit data using 8086 processor.

```

DATA SEGMENT
NUM1 DW 0005H
NUM2 DW 0004H
DATA ENDS

CODE SEGMENT
ASSUME CS:CODE, DS:DATA

START:
MOV DS, DATA
MOV AX, NUM1
MUL NUM2
MOV RESULT, AX
INT 04Ch

```

The screenshot shows the emu8086 interface with the assembly code loaded. The registers window shows the state of the 8086 registers. The stack window shows the current stack pointer. The status bar at the bottom indicates the current line of code is 6 and the column is 26.

12. Write an assembly language program for dividing two 16-bit data using 8086 processor.

```

DATA SEGMENT
NUM1 DW 0014H
NUM2 DW 0004H
DATA ENDS

CODE SEGMENT
ASSUME CS:CODE, DS:DATA

START:
MOV DS, DATA
MOV AX, NUM1
DIV NUM2
MOV QUOT, AX
INT 04Ch

```

The screenshot shows the emu8086 interface with the assembly code loaded. The registers window shows the state of the 8086 registers. The stack window shows the current stack pointer. The status bar at the bottom indicates the current line of code is 24 and the column is 10.